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Perform a complete EDA on the `penguins` dataset (already loaded) and answer the following in new cells below. This is meant to be done **independently** — use everything you've learned in this notebook!

1. Number of rows and columns
2. Number of missing values (per column)
3. Number of duplicate rows
4. A correlation matrix / heatmap of the numeric columns
5. A boxplot showing `bill_length_mm` grouped by `species`
6. A pairplot colored by `species`
7. A fully cleaned version of the dataset (missing values handled, duplicates removed)

```
import seaborn as sns
penguins = sns.load_dataset("penguins")

# Display the number of rows and columns in the dataset

print("Shape of Dataset:", penguins.shape)

print("Rows:", penguins.shape[0])
print("Columns:", penguins.shape[1])
```

```
Shape of Dataset: (344, 7)
Rows: 344
Columns: 7
```

```
# Count missing values in each column

print("Missing Values Per Column:")

penguins.isnull().sum()
```

```
Missing Values Per Column:
```

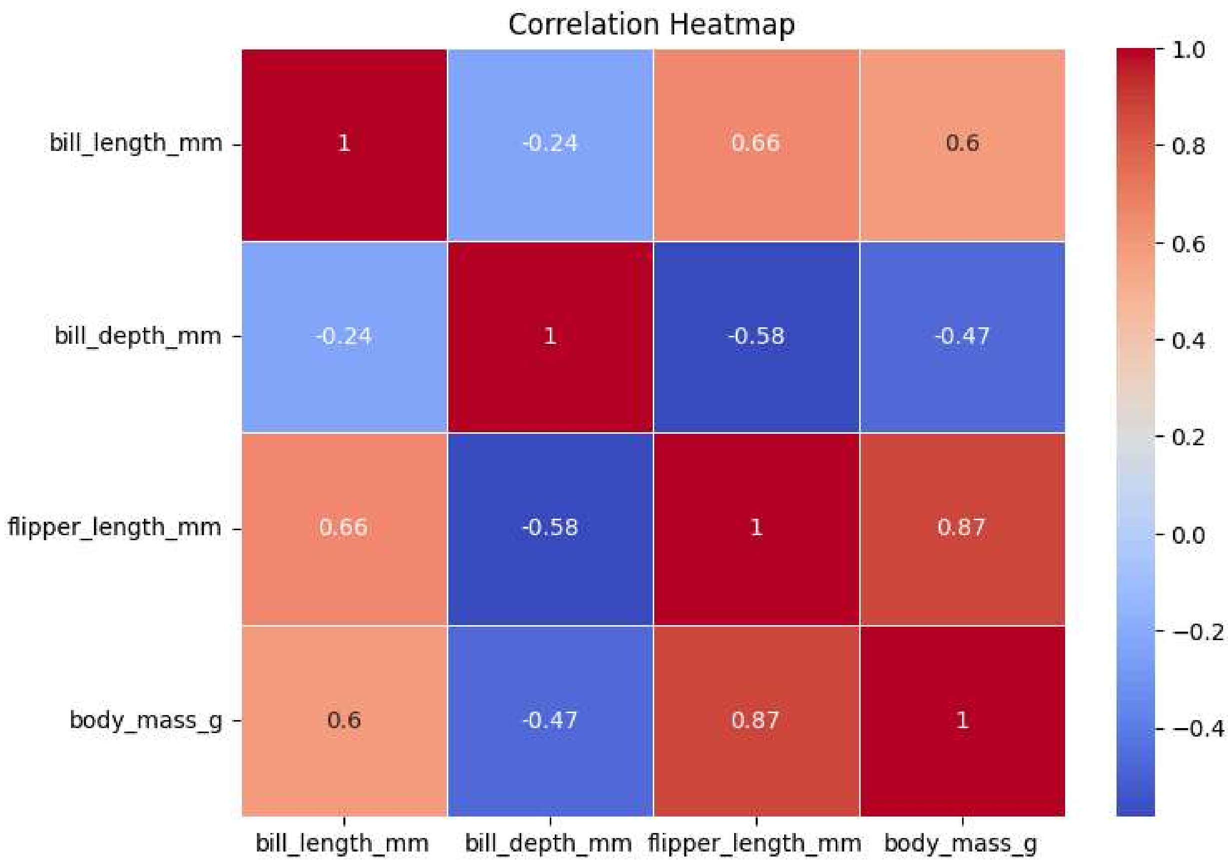
| | 0 |
|---|---|
| **species** | 0 |
| **island** | 0 |
| **bill_length_mm** | 2 |
| **bill_depth_mm** | 2 |
| **flipper_length_mm** | 2 |
| **body_mass_g** | 2 |
| **sex** | 11 |

**dtype:** int64

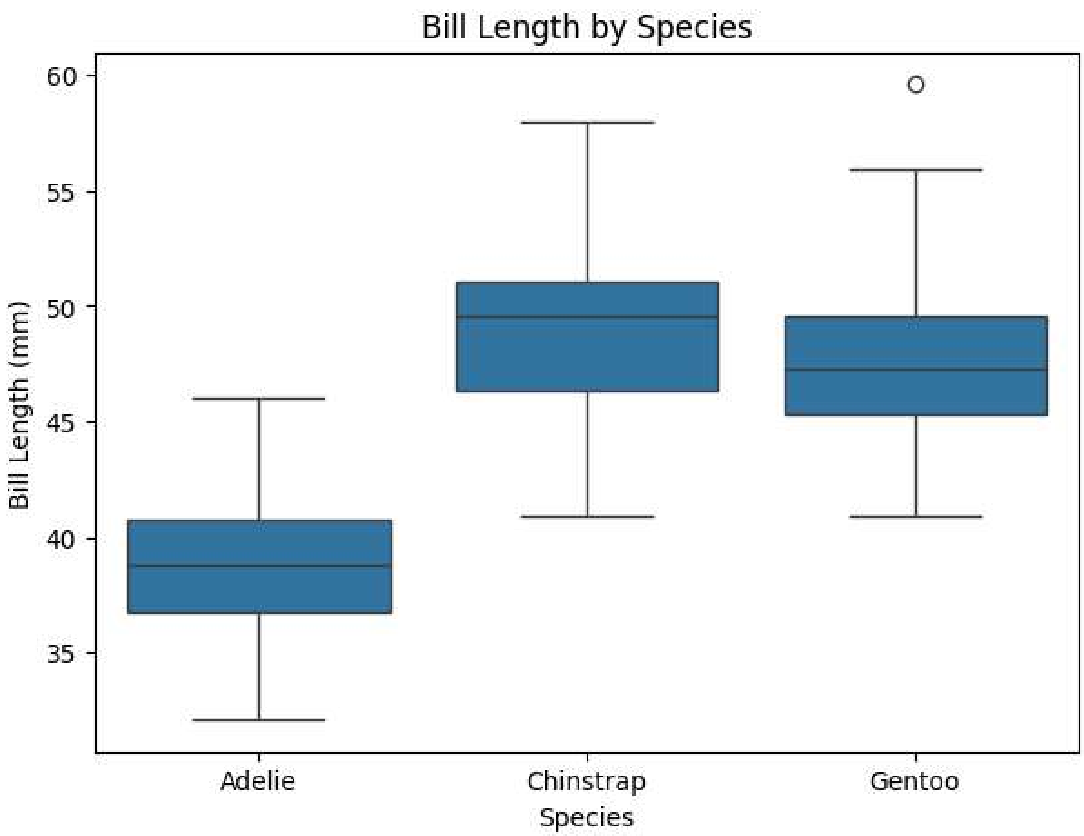
```
# Count duplicate rows

duplicates = penguins.duplicated().sum()

print("Number of Duplicate Rows:", duplicates)
```

```
Number of Duplicate Rows: 0
```

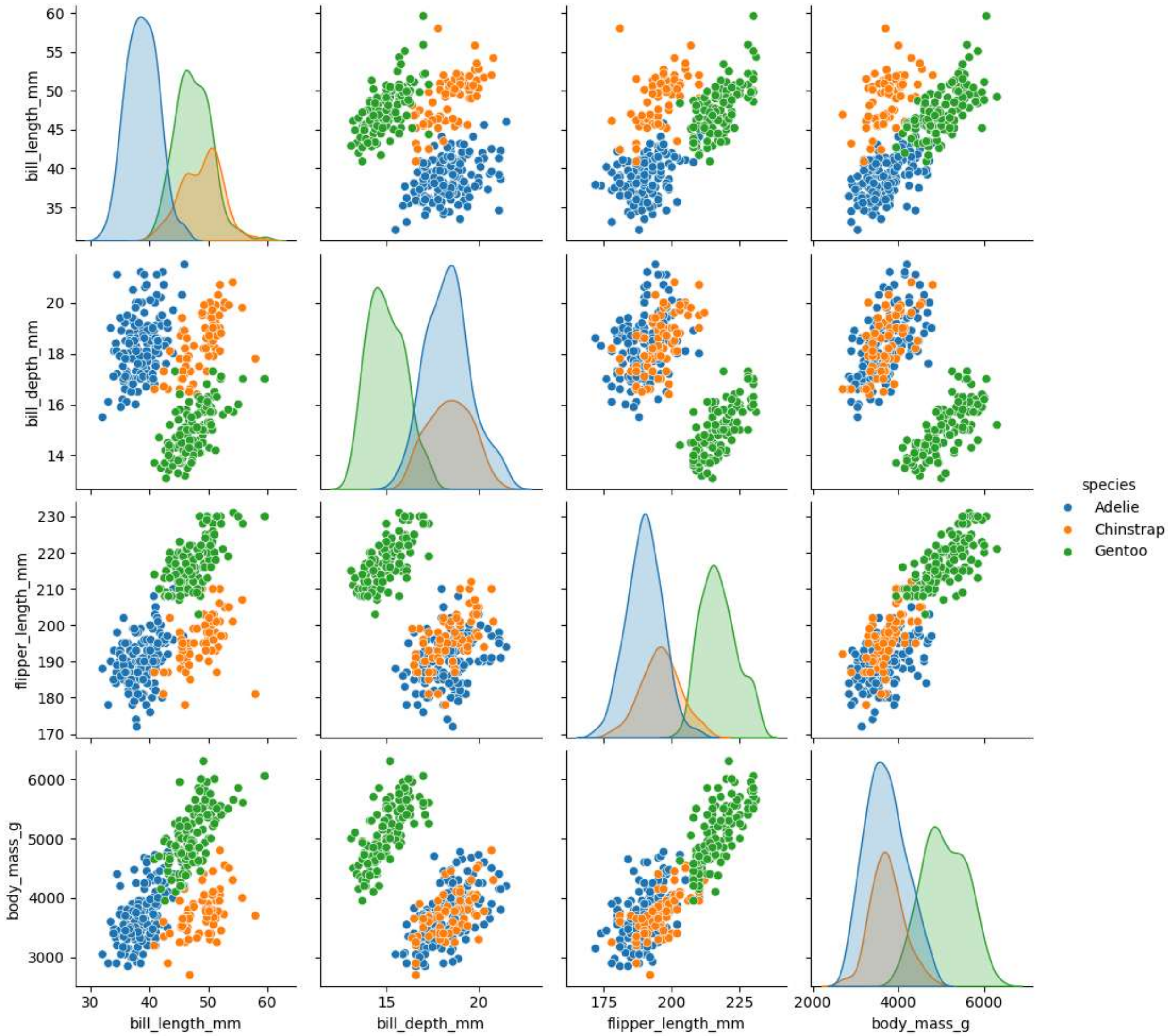
```
# Import required libraries

import seaborn as sns
import matplotlib.pyplot as plt

# Select only numeric columns
numeric_data = penguins.select_dtypes(include="number")

# Calculate correlation
corr = numeric_data.corr()

# Display heatmap
plt.figure(figsize=(8,6))

sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    linewidths=0.5
)

plt.title("Correlation Heatmap")
plt.show()
```




```
# Create a boxplot

plt.figure(figsize=(7,5))

sns.boxplot(
    x="species",
    y="bill_length_mm",
    data=penguins
)

plt.title("Bill Length by Species")
plt.xlabel("Species")
plt.ylabel("Bill Length (mm)")

plt.show()
```




```
# Pairplot of numeric variables

sns.pairplot(
    penguins,
    hue="species"
)

plt.show()
```




```
# Create a copy so the original dataset remains unchanged

penguins_clean = penguins.copy()

# Remove duplicate rows
penguins_clean = penguins_clean.drop_duplicates()

# Remove rows containing missing values
penguins_clean = penguins_clean.dropna()

# Display information about cleaned dataset
print("Shape of Cleaned Dataset:", penguins_clean.shape)

# Display first five rows
penguins_clean.head()
```

```
Shape of Cleaned Dataset: (333, 7)
```

| | species | island | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | se |
|---|---|---|---|---|---|---|---|